INCH-POUND

MIL-S-70547 (AR)
AMENDMENT 7
26 June 1998
SUPERSEDING
AMENDMENT 6
04 October 1994

MILITARY SPECIFICATION

SPHEROIDAL PROPELLANT, M38 FOR USE IN
PROPELLING CHARGES M218, M219 AND
XM220 FOR CARTRIDGES, 81MM SMOKE, M819,
ILLUMINATING M853 AND PRACTICE, XM879

This amendment forms a part of MIL-S-70547 (AR), dated 5 September 1986, and is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add “(M220)” and “(M879)”. Make these changes whenever reference to XM220 and XM879 appears in the specification.

* 2.1.1, Specifications, Military: Add the following new specifications:

“MIL-P-3984 – Propellants For Small Arms Ammunition”

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* 2.2 Other publications: Delete “49 CFR – Interstate Commerce Commission Rules and Regulations For The Transportation of Explosives and Other Dangerous Articles.”

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3.2.1. Chemical composition:

Table I Composition: Delete “0.85 plus/minus 0.25” and substitute “0.85 to 1.50”.

3.4 Table II, III and IV: Change Individual Chamber Pressure (Max) at 145 degrees F from “13,100” to “14,500”.

Table III: Change Muzzle Velocity (Mean) at 70 degrees F from “1020 ft/sec” to “1013 ft/sec”.

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* 3.2.1:

After “Potassium Sulfate” add:

“Calcium Carbonate … 0.1 to 0.4 … 4.5.1”

Total Volatiles: Delete “2.0” and substitute “3.5”

Residual Solvent: Delete “1.20” and substitute “2.50”

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* 3.6 Delete “Calcium Carbonate … 0.25 … 4.5.1”

PAGE 7

Delete in its entirety and substitute new paragraph 4.4.1.3:

“4.4.1.3 Ballistic performance. Five containers shall be selected at random from each lot of propellant. Equivalent quantities of the five containers shall be taken such that the aggregate quantity shall be sufficient for ballistic performance tests described in 4.5.3. The propellants shall be aged a minimum of forty-

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five (45) days at sealed and ambient conditions from the date of the packing (unless other conditions specified), prior to ballistic testing."

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* 5.1.3: Delete in its entirety and substitute new paragraph 5.1.3:

"5.1.3 Level C. Pack in accordance with Level C requirements of MIL-P-3984".

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* 5.2.3 Delete in its entirety and substitute new paragraph 5.2.3:

"5.2.3 Level C. Marking shall be in accordance with the Level C marking requirements of MIL-P-3984 with the following exception. Under special marking instructions, section 5.2.3 (a) the label shall read as follows:

AFTER TWO YEARS FROM DATE OF MANUFACTURE, APPROVAL BY THE RESPONSIBLE PROCURING AGENCY IS REQUIRED PRIOR TO THE LOADING OF THIS PROPELLANT INTO MORTAR PROPELLING CHARGES."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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